

Sumith Raj

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Summary

- Big Data Engineer and Developer with over 5 years of industry experience in application development and Engineering.
- Proficiency in Python, Scala, and SQL with extensive experience in cloud computing platforms such as Azure and AWS.

EXPERIENCE

Big Data Engineer, Conduent, NJ

Feb 23 – Present

- Developed a secure, scalable data pipeline using Data Factory that triggers Azure functions in response to blob storage.
- Additionally, I formulated efficient Oozie workflows for Medicaid program data, aligning with Conduent Medicaid Suite (CMdS) requirements. Utilized compression codecs in HDFS for sizeable medical bill review datasets, potentially achieving up to 40% faster processing and cost reduction.
- Built the Python packages for data pipelines within Airflow DAG, rigorously testing across Development and Production environments, and was assigned to implement database architecture, data modeling, and schema validation.
- Developed and Implemented a near real-time data pipeline using spark structured streaming, extracting data from Cassandra and incremental importing it into HDFS, enabling timely access without data loss for critical operations.
- Architected a highly reliable data pipeline with 94.7% uptime, ingesting 65 GB transactional data (EHR) from diverse sources using Apache Spark. This solution ensures efficient data processing for analysis and improved patient care.

Data Engineer, U.S. Bank

Aug 2022 – Dec 2022

- Automated data cleansing and standardization for large datasets in Azure Blob Storage using a Python script within Azure Function, significantly improving data quality by 60% and reducing manual effort.
- I also designed and implemented a secure data lake solution on Azure Data Lake Storage (ADLS) for customer financial data, focusing strongly on data governance and privacy regulations.
- Migrated on-premises systems to a containerized architecture using Azure Kubernetes Service (AKS), resulting in a 30% reduction in operational costs.
- Established Azure Stream Analytics to process streaming data in real-time and store it in Blob storage for further analysis and Machine Learning training.

Technical Data Analyst, IKS Health, Hyderabad.

Aug 2017 – June 2021

- Developed ETL jobs to load fixed-width files from the mainframe server into Redshift using batch jobs.
- I've used Python to read real-time data from the Streaming Data Platform (SDP) and load it into the SAP Analytic Data Warehouse cloud.
- Build and automate AWS Glue jobs to ingest data coming from SFTP vendor systems. Implemented a Crawler and catalog tables to manage the Schema changes of incoming files automatically.
- Leveraged SAS to analyze hospital readmission rates and identify high-risk patient populations, leading to a 10% reduction in readmission rates.
- Developed proof-of-concept (POC) to test building batch jobs with dependencies using AWS Batch. Jobs are triggered using a dependencies JSON input Script initiated by the process using Airflow
- Implemented a machine learning infrastructure for a medical manufacturing client to improve product quality.
- Created UNIX shell scripts to load files into HDFS from the Linux File System, tuning query performance and optimizing Cronjobs and shell scripts.

CERTIFICATIONS

- (Amazon) AWS Certified Solutions Architect – Associate.
- (Microsoft) Microsoft Certified: Azure Data Engineer Associate.
- (IBM) ETL and Data Pipelines with Shell, Airflow, and Kafka.

EDUCATION

Master of Science, Computer Systems and Analytics

Aug 2021 - Dec 2022

University of Bridgeport

GPA: 3.7

Bachelor of Technology, Computer Engineering

June 2013 - May 2017

Osmania University

GPA: 3.4

SKILLS

Programming Languages: Scala(Circe), Python(NumPy, Pandas, Matplotlib), ML(TensorFlow, Scikit-learn).

Databases/Data warehousing: MySQL, NoSQL(MongoDB, Cassandra, DynamoDB), PostgreSQL, Redshift, Snowflake.

Cloud Providers: AWS(EC2, S3, Lambda, Redshift, Kinesis), Azure(Data Factory, Blob Storage, Stream Analytics).

Big Data Technologies: Apache Spark, Hadoop, Apache Kafka.

ETL & Orchestration Tools: Apache Airflow, AWS Glue, Azure Kubernetes Service (AKS).

Data Visualization: Tableau, Power BI.

Version Control: GitHub, Bitbucket, GitLab.